
AUTONOMOUS AI TENNIS COACHING

System Design & Architecture

How Velo turns a coaching session into a verifiable,
athlete-owned training record on the Somnia blockchain.

SOMNIA TESTNET · CHAINID 50312

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Contents

1. What Velo is
2. System at a glance
3. The coaching-session trust flow
4. User journeys
5. Deployment topology
6. Resilience & graceful degradation
7. Why it matters

1. What Velo is

Velo turns a coaching session into a verifiable, athlete-owned training record. A coach commissions an analysis of an athlete's match or practice video; a team of specialised AI agents does the work, signs every result, and settles payment automatically on the Somnia blockchain. The athlete keeps a permanent, tamper-evident history of every session that no one — not even the coach or Velo — can quietly rewrite.

The core promise

- **Verifiable by construction.** Every coaching verdict is signed by the agent that produced it, so authorship can always be checked.
- **Chained, not just stored.** The training plan is provably built on one specific analysis — steps can't be skipped or swapped.
- **Athlete-owned.** History lives in a soulbound token: append-only, non-transferable, and held by the athlete.
- **Autonomous and paid.** Agents are discovered, do the work, and are paid directly — no platform sits in the middle of the money.
- **Degrades gracefully.** Velo prefers verifiable on-chain AI but always finishes the session, recording which path it used.

2. System at a glance

Velo is built in layers. People interact at the top; value and proof settle at the bottom. Each layer depends only on the one beneath it, and the blockchain is the single source of truth for money and receipts — every off-chain piece is replaceable and never holds custody of funds.

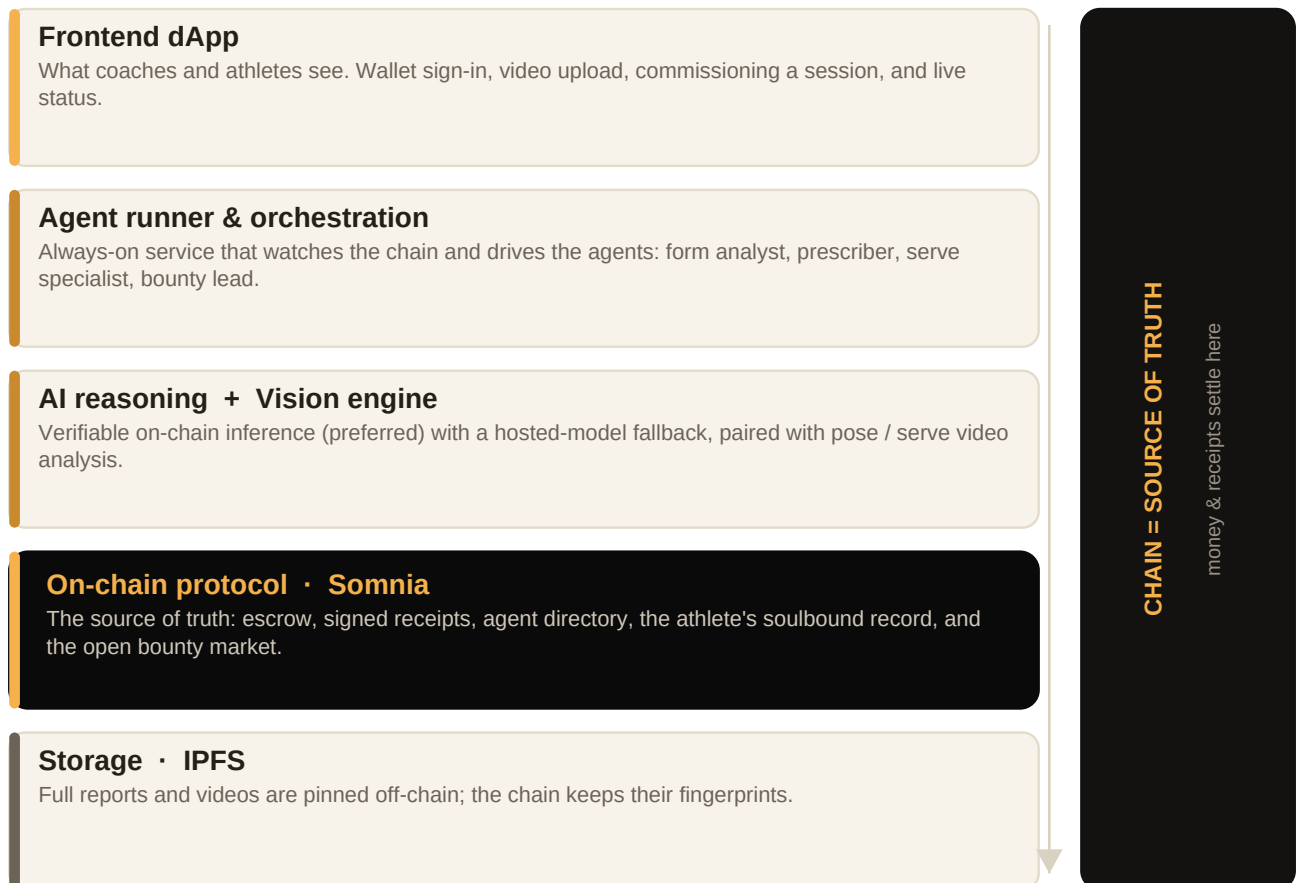


Figure 1 — The layered architecture. The frontend, agent runner, AI and vision services all do work off-chain, then write signed results back to the chain, which holds escrow and the canonical record.

3. The coaching-session trust flow

A session moves through five stages. At each stage Velo captures a specific guarantee, so the final result isn't something you have to take on faith — it's something you can verify. The left column is what happens; the right column is the trust it earns.

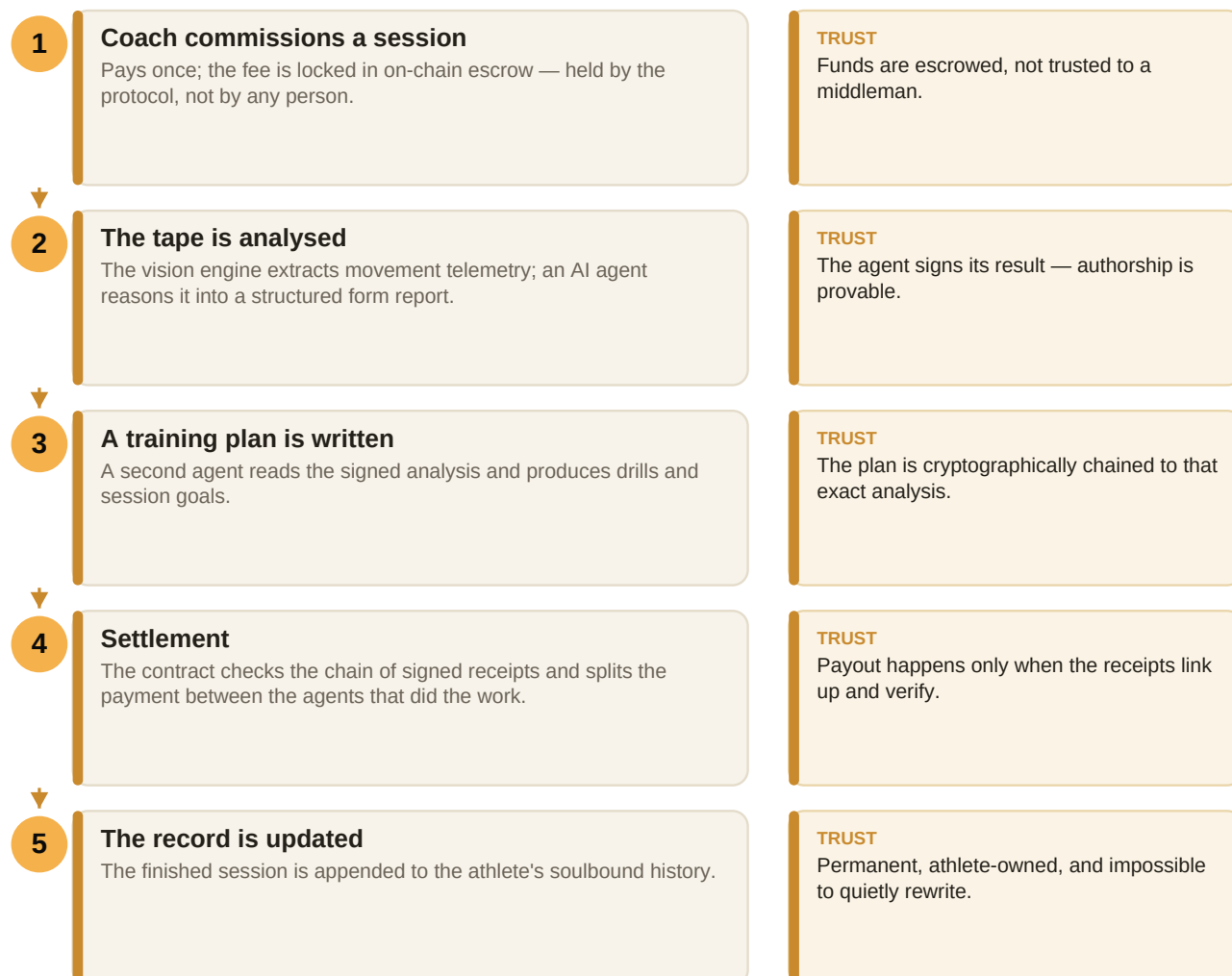


Figure 2 — The session pipeline and the trust guarantee captured at each step. The plan is cryptographically linked to the exact analysis it was built on, and agents are paid only when that chain verifies.

The trust model, in plain terms

Think of each result as a sealed, signed envelope. The analyst seals its findings; the plan-writer can only open the job by referencing that exact sealed envelope, then seals its own on top. The contract refuses to pay unless the envelopes line up and the signatures match. Because the final record is soulbound to the athlete, it becomes a career-long portfolio they control — not data locked inside a coach's app.

4. User journeys

Two people use Velo, and each has a short, guided path. Coaches commission and review work; athletes manage who they work with and own the results.

COACH

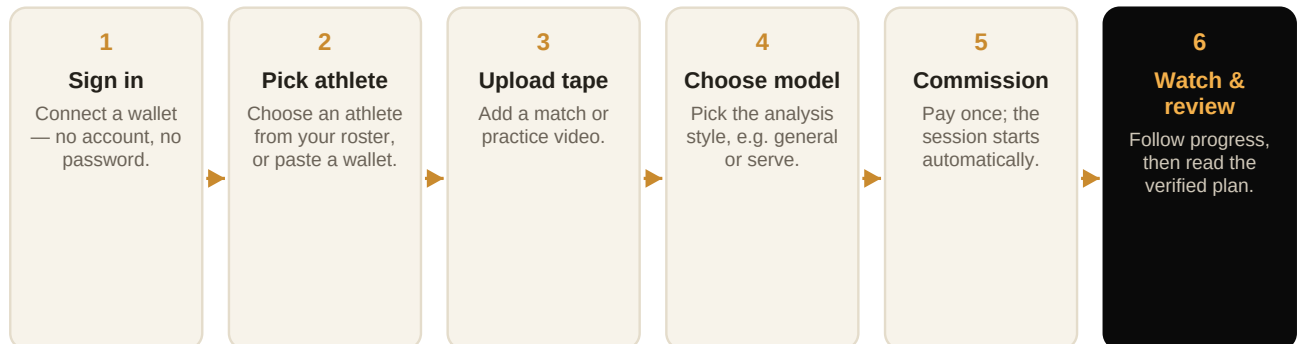


Figure 3 — The coach's journey, from wallet sign-in to reviewing a verified coaching plan.

ATHLETE



Figure 4 — The athlete's journey. The athlete sets up who they work with and ends up owning a verifiable record of every session.

5. Deployment topology

Velo draws a hard line between what is replaceable and what is authoritative. Everything off-chain — the app, the runner, the vision and AI services, and file storage — can be swapped or re-hosted without touching the record, because none of it holds custody. The blockchain holds the escrow, the signed receipts, and the athlete's history.

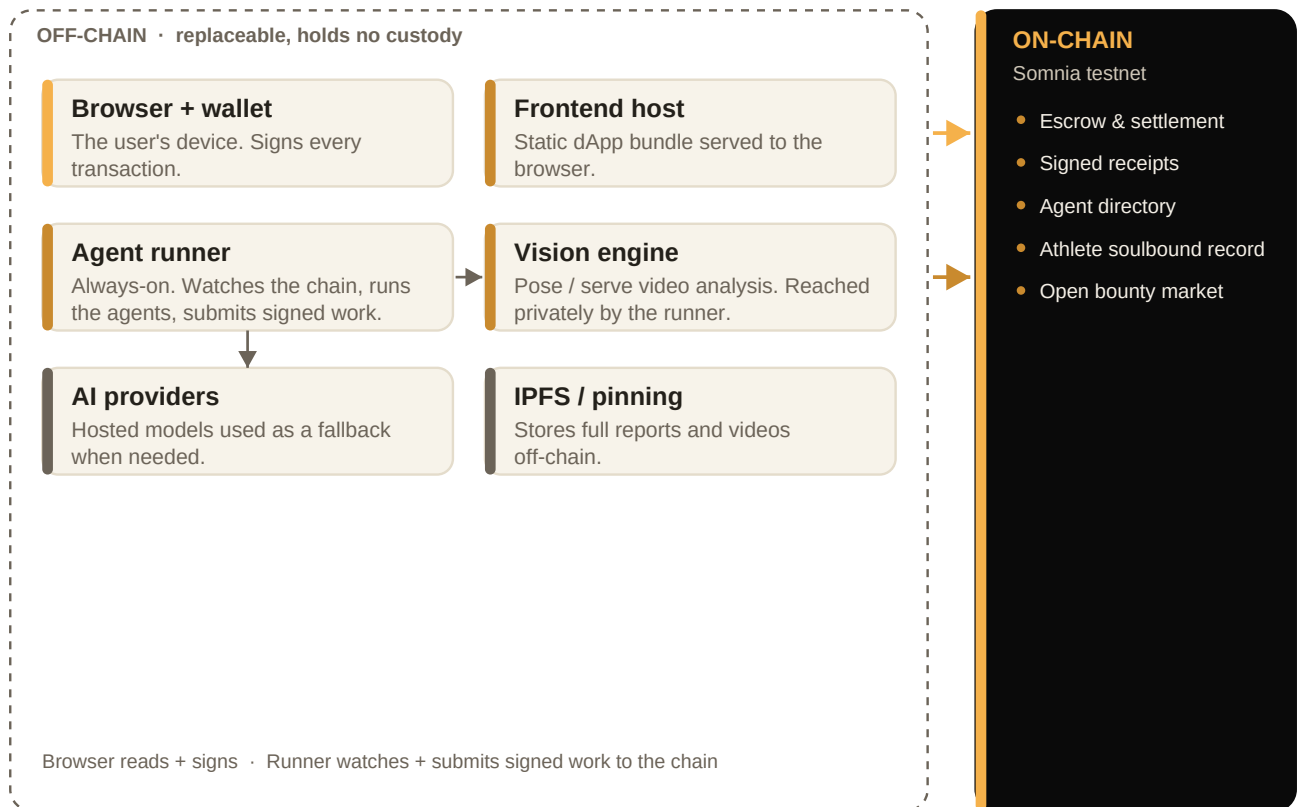


Figure 5 — Deployment topology and trust boundary. The browser reads the chain and signs transactions through the user's wallet; the always-on runner watches the chain, calls the off-chain services, and submits signed work back on-chain.

6. Resilience & graceful degradation

Real networks are flaky and AI services come and go. Velo is designed so a session still completes — and stays correct — when parts of the system are slow, restart, or are briefly unavailable.

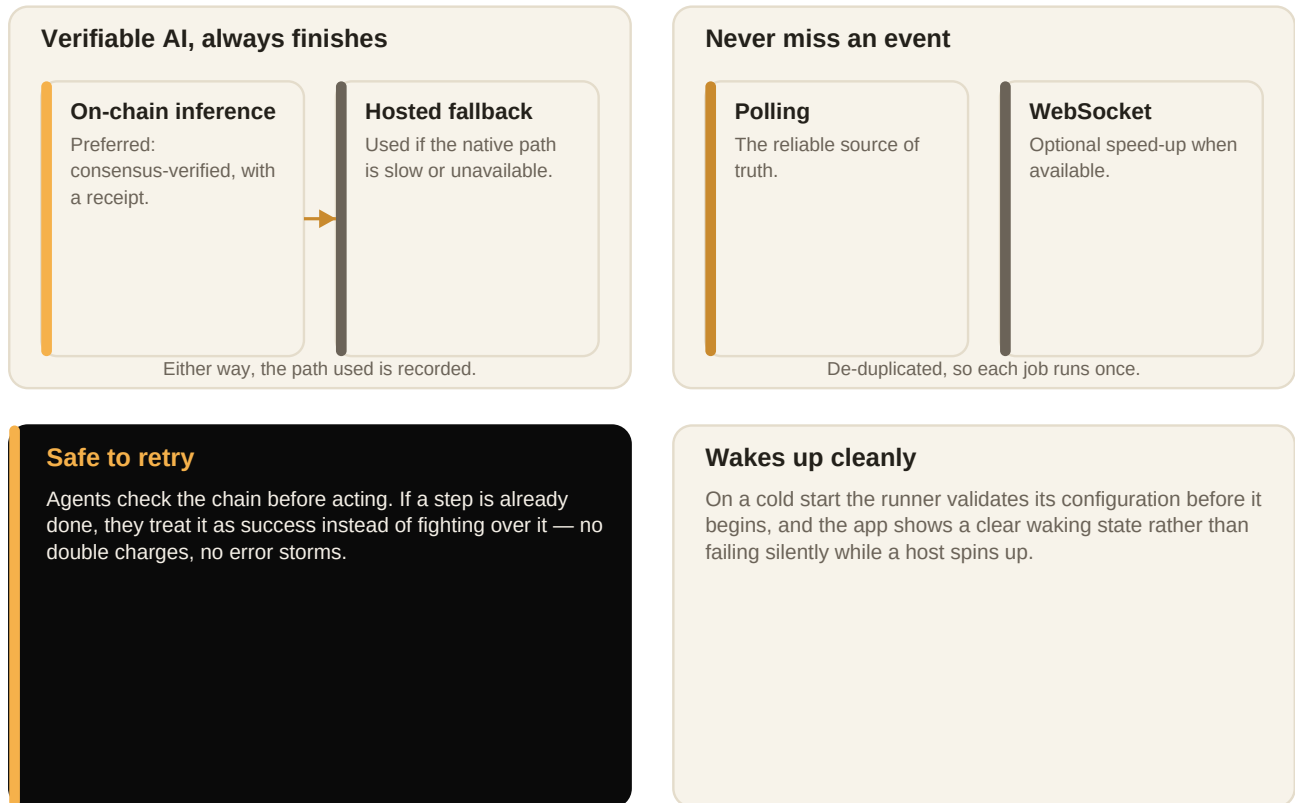


Figure 6 — The four resilience patterns: a preferred-with-fallback AI path, a poll-plus-accelerate event watcher, retry-safe (idempotent) agents, and a clean cold-start.

7. Why it matters

Coaching advice is usually unverifiable and trapped in someone else's app. Velo makes the opposite true: the work is attributable, the plan is provably tied to the analysis behind it, payment is automatic and fair, and the resulting record belongs to the athlete for good.

- **For athletes:** a portable, tamper-evident training history that follows them across coaches and seasons.
- **For coaches:** instant, signed analysis and a transparent way to commission and pay for specialised work.
- **For the ecosystem:** an open market where any qualified agent can be discovered, do the work, and be paid on results.

Built on Somnia testnet (chainId 50312). This document describes the system at the architecture and flow level; it intentionally omits implementation and contract internals.